369 Final Project Analysis Draft

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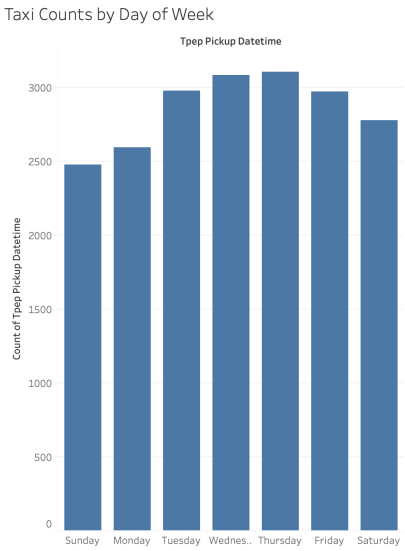
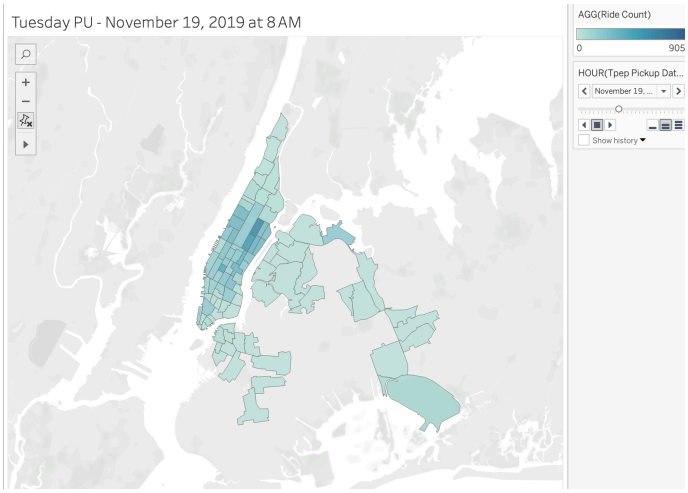
For our Intro to Distributed Computing final project, I'm using a dataset that includes every taxi ride taken in New York City in 2019. The set includes taxi ride pickup and dropoff region, trip duration, trip distance, fare, passenger tip, and passenger count. Pickup and dropoff locations are one of 255 numbered taxi regions within the city. This dataset is enough to give us a comprehensive understanding of taxi behavior and patterns in the city. For this project, I'm hoping to understand how external factors change taxi demand and usage patterns across New York City. For the sake of scope, we will focus on how holidays affect taxi usage patterns in NYC.

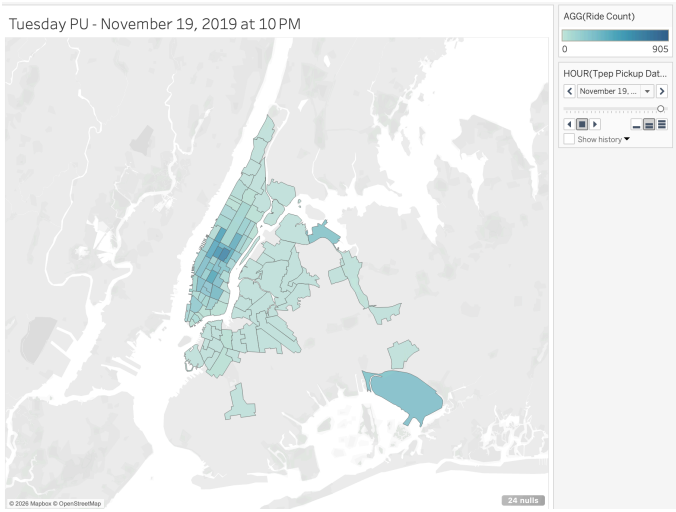
To identify abnormal behaviors, we'll first need to look at baseline taxi patterns. The graph to the right outlines average taxi ride counts by day of the week. We can see that each week, ridership increases throughout the week from Sunday, peaking on Thursday. This graph was made using a random sample of 400k taxi rides. Actual ride counts in the graph are inaccurate, but day-by-day relative counts are significant.



To further understand baseline usage, below are taxi zone pickup counts for a typical Tuesday, 11/19.



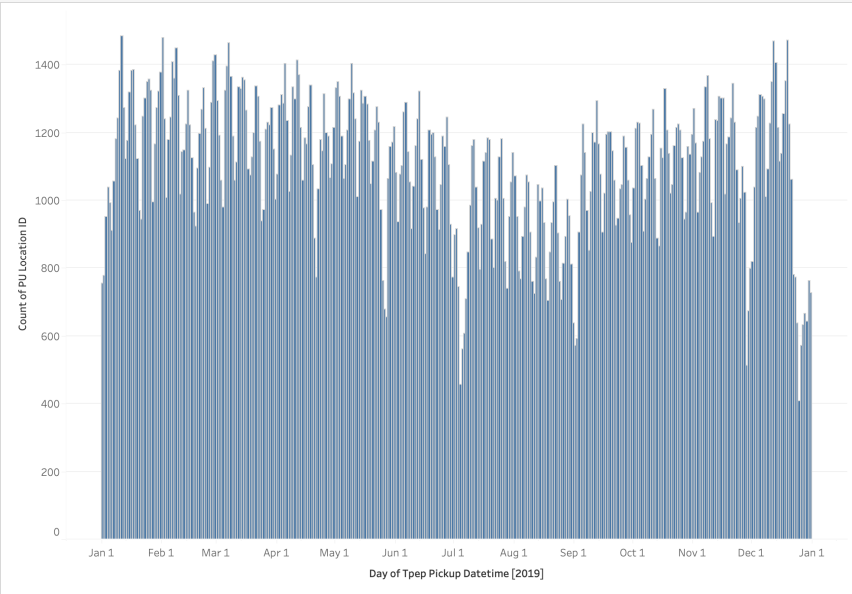
Tuesday Ride pickups 8-9am



Tuesday Ride dropoffs 10-11pm

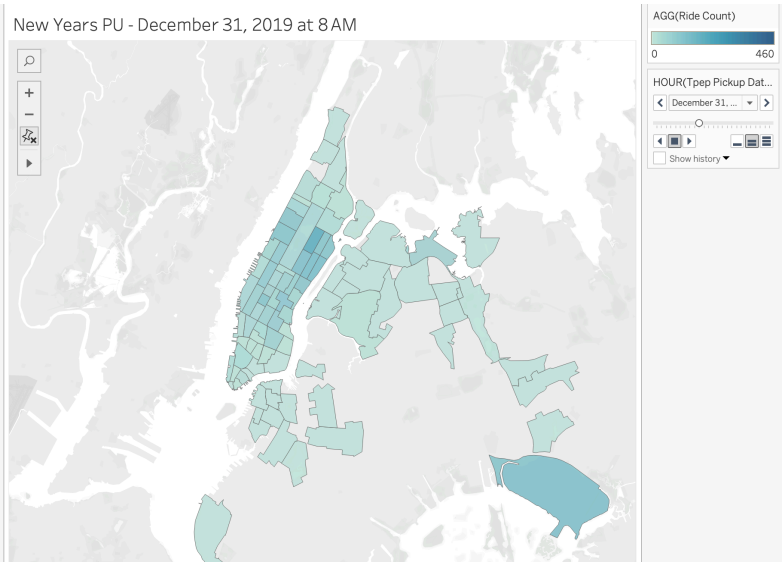
Throughout the day, there is a general shift in ride behavior. In the morning there is a concentration of pickups from the upper east side and its adjacent neighborhoods. In the evening, pickups are concentrated more strongly further south around Times Square.

As for abnormal behavior, graphing ridership counts at day-level precision nicely illustrates the effects that external factors can have on ridership. The graph to the right shows ridership throughout the year. We can see the weekly signal of rider behavior, but there are other factors at play too. General ridership appears to fluctuate seasonally as well. There is also severely reduced ridership on certain days, including the 4th of July and New Year's Eve

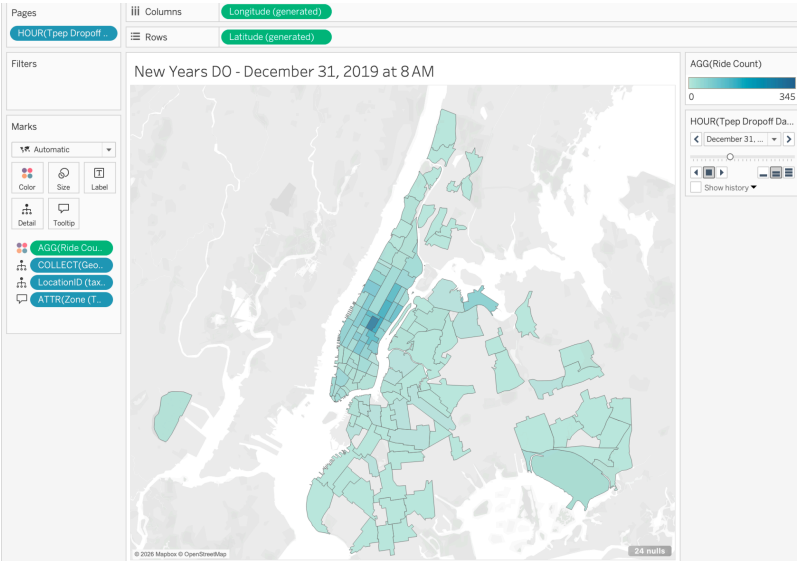


Ride counts day by day in 2019 from random sample of 400k rides.

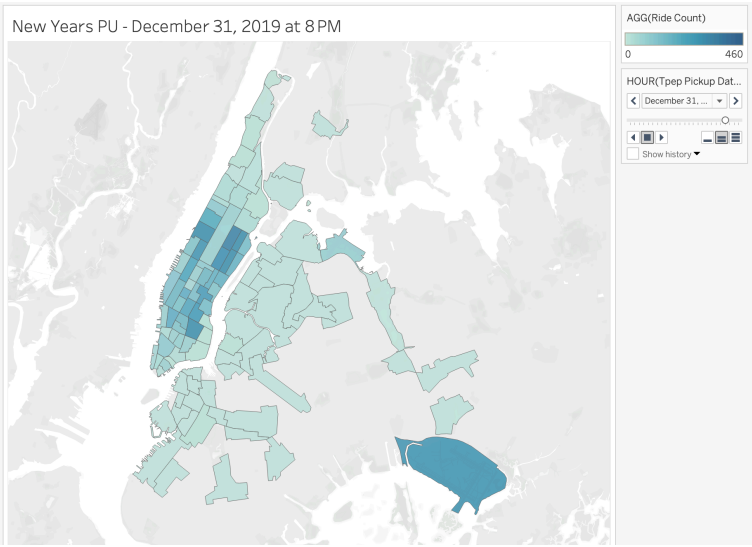
We'll dive deeper into New Year's Eve. The following graphs are pickup and dropoff locations sampled throughout the day on 31 December 2019.



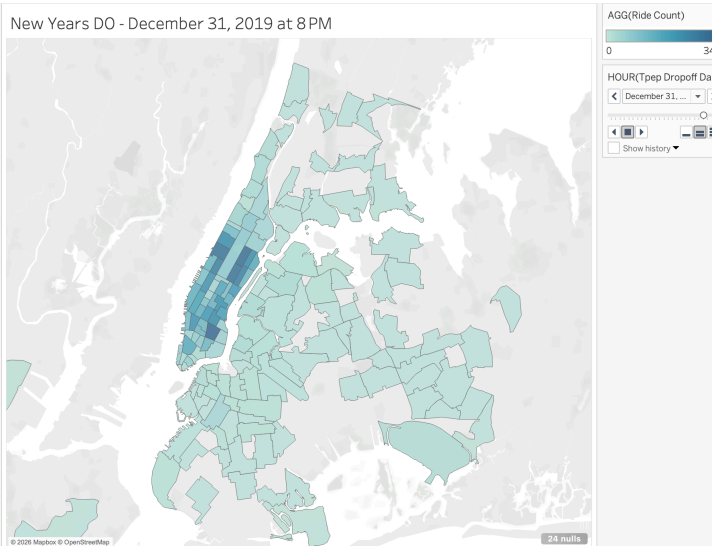
Ride pickups 8-9am NYE



Ride dropoffs 8-9am NYE



Ride pickups 8-9pm on NYE



Ride dropoffs 8-9pm on NYE

We see significantly different taxi behavior on New Year's Eve. Total ridership throughout the city is significantly lower than any typical non-holiday. In the morning, many pickups are spread around Manhattan, with the Upper East Side receiving the most in the region. La Guardia, and especially JFK airport see some traffic, but not as much as typical. For dropoffs in this timeframe, the Midtown Center zone, right next to Time Square, sees a drastically higher count than normal. Even on the morning of New Year's Eve, many people are concentrating at the celebration site for that night.

Taxi behavior becomes more interesting at night on New Year's Eve. From 8-9pm we see a bulk of traffic around Manhattan, but also a large number of rides from JFK and La Guardia. A number of people are flying into the city just before the New Year's Celebration.